

When Does Personalization Earn the Route?

A Full-Catalog Temporal Study of Guarded Recommendation

Ahnaf An Nafee

23 September 2026

Abstract

Offline recommendation gains can disappear when a model must retrieve from the full catalog rather than reorder previously observed items. We study a practical decision: whether a personalized challenger should replace a validation-selected popularity baseline. Using the global temporal splits of Amazon Reviews’23, we train popularity and co-review models only on earlier reviews, retain future targets absent from the training catalog as misses, and fix the route before reading test reviews. Video Games is an exploratory development category; Musical Instruments is a previously unopened category evaluated after the method was fixed. On 35,905 eligible Musical Instruments test requests, a co-review hybrid obtains 0.008681 NDCG@10 against 0.008007 for recent popularity. The paired difference is 0.000674, with a user-cluster bootstrap 95% interval of [0.000040, 0.001338]. A later two-tower retriever improves on popularity when blended with it, but does not beat the active hybrid on validation and remains a shadow model. We then audit what the frozen route leaves unread: low star ratings, the age and grade of prior positive reviews, and history length. A repulsion arm and an age-decay arm improve on the active route by 0.000449 and 0.000466 NDCG@10 on the same validation cohort used to select them. These descriptive intervals do not establish a fresh gain. A first-positive-review prior loses, while a simulated interview is reported separately as a hindsight-oracle sensitivity analysis, not observed user behavior. A final arm answers a typed sentence against product listings; it is served, but on the full test split of both categories, read after its configuration was fixed on validation, it does not clear the gate on the frozen metric, and a reach decomposition explains the direction; the encoded variant that re-scores the same shortlist is directionally positive on the validation cohort, but on the full Musical Instruments test split it loses and does not open the gate. These latter experiments are exploratory, and the profile arms are scored on the validation cohort only. A local service exposes the chosen route, shadow ranking, and fallback behavior. The contribution is an auditable comparison and a guarded routing implementation, not a new recommendation algorithm or evidence of online impact.

Keywords: recommender systems; full-catalog evaluation; temporal split; model selection; neural retrieval; fallback routing; already-held evidence; preference elicitation; phrase-to-products retrieval

1 Introduction

Predicting a rating for an item a user chose to review is different from retrieving that item among thousands of products. The second task must account for products the user has not encountered, items unavailable to a model trained on earlier activity, and changes in the active population. Offline protocols that ignore the global timeline can distort model comparisons [3, 4]. We therefore evaluate retrieval against the training catalog under a global temporal split.

The operational question is narrower than finding the strongest model in a benchmark: when does a personalized challenger have enough validation evidence to replace a simple route? We compare a train-only cosine co-review model blended with popularity against the better of lifetime and recent popularity. We then evaluate a compact two-tower retriever as a later challenger. Both personalized methods use established techniques. What matters in this study is that the candidate set, temporal boundary, selection rule, and fallback decision are visible together, including a case where the more complex model is declined.

The held-out Musical Instruments result is positive but small in absolute terms. It is therefore useful as evidence that the fixed co-review route can transfer to another product category under the same benchmark, rather than evidence of a broadly superior recommender. The neural extension, designed after that category’s test result was known, provides an exploratory model comparison and a concrete example of a validation gate remaining closed.

Two further sets of experiments ask a different question about the same frozen route: what does it leave unread? The route reads one field of a request: the last twenty reviewed items, including poorly rated ones, with no star information attached to the scorer. The archive also holds ratings, review times, older items, and requests with no history to read. Arms that re-read existing ratings or review times require no new prompt; a simulated questioning arm is reported with its question count and the answers recovered by a hindsight oracle. A final arm supplies a capability the route lacks entirely, answering a sentence about what the person wants. Every one of these challengers is judged by the rule the route itself was selected under, and more than one of them is declined.

2 Related work

Global temporal evaluation is important when future interactions or future catalog information could reach training or selection [3, 4]. We use the owner-provided time split of Amazon Reviews’23 [2, 1] and freeze model fitting before the validation origin. The split rule is not a neutral detail: it can markedly alter the ranking of published models, and temporal global splitting is recommended as the default [21], while sampled test items and overlapping periods are among the flaws that can reverse a reported conclusion [22]. The benchmark’s 5-core filter is itself retrospective, however, so this procedure does not recreate an unfiltered production stream.

Item co-occurrence, popularity controls, and two-tower retrieval are established recommendation tools. Deep candidate-generation systems have demonstrated the utility of learned user and item representations [5]; in-batch alternatives are a common training device for two-tower models [6]. Our neural experiment uses those ideas without claiming a novel architecture or negative-sampling method. The paper instead reports whether the trained challenger improves the route already selected by a simpler model.

A route that is allowed to decline has other names in the literature. Serving systems place a latency objective between callers and models and answer heavy load by caching, batching, or switching to a cheaper model [8]. Learning to abstain treats rejection as a decision in its own right, measured separately from predictive quality and separated into ambiguity and novelty cases [9], and learning when to defer to a second decision-maker requires evidence about that decision-maker’s actual capability [10]. The gate in this paper is an offline, uncalibrated version of the same decision, kept in code rather than in a learned policy: it continues to score the declined route so that evidence for reopening it accumulates, and it returns a named reason instead of a silent substitution.

That a challenger beats another challenger is a deliberately weak warrant here. Offline metrics correlate with online performance, but training and interaction dynamics weaken the relationship [11], and counterfactual estimates of uplift from logs trade bias against variance in precisely the personalised setting [12]. Our intervals are therefore read as statements about a frozen cohort and temporal period, not as expected traffic movement. Published guidance for this literature asks for baselines that are tuned rather than assumed weak, a stated search budget, and explicit uncertainty quantification, and observes that many published results on smaller datasets are probably not statistically significant once uncertainty is quantified by nested resampling and testing [23]. Our search budget is small and fixed: four blend weights for the co-review route and five for the neural route, with no other tuning. The baseline is the route that same selection rule already chose, and every difference we report carries a paired user-cluster interval.

The two remaining challenger families each have prior evidence. Evidence a request omits can be

asked for: preference elicitation has long been studied for new users with no history to personalize from [7]; recent work evaluates onboarding elicitation policies against simulated users instead of live traffic [13] and finds that the elicitation interaction itself introduces selection bias [14]. Both are reasons to publish a question count beside a gain rather than a gain alone. Evidence can also be read from wording. Embedding-based retrieval has been applied to large-scale search [15], the ESCI benchmark supplies product-query relevance judgments [16], and sentence embeddings permit product listings to be encoded before a query arrives [17]. Because the phrases in our language arm are constructed rather than typed, that arm inherits a measurement result from synthetic test collections: bias can affect absolute metrics, while comparisons between systems may be less affected [18, 19]. This motivates a paired comparison, but does not validate our constructed phrases as real user requests. A production RAG study of routing makes the same point from the traffic side: synthetic queries implied the expensive path was needed for over 90% of requests where real traffic needed it for 27.8% [20].

Two published lines sit closer to the arms that opened the gate. The first reads what a person registers without recording it: short-video logs where the absence of an expected action is modelled as passive-negative feedback [24], and sequential recommenders scored with explicit negative feedback against a likes-only baseline [25]. Both show why negative feedback can be useful, but skips of auto-played videos and explicit negative actions differ from a low rating on a product the person chose to review. The second line concerns the fallback. A re-examination of the popularity baseline finds that for heavy users the individual history predicts later engagement better than popularity does [26]. Our size-conditioned arm selects more popularity weight for long histories on this validation cohort. Because its bucket weights were also chosen there, the pattern is exploratory rather than a contradiction of the MovieLens result.

3 Evaluation design

3.1 Data, target, and catalog

We use the `timestamp_w_his` train, validation, and test files for Video Games and Musical Instruments in the Amazon Reviews’23 5-core benchmark [1]. The source supplies global cutoffs and each review’s prior-item history. A rating of at least four defines a positive review. Each eligible positive validation or test review becomes one request whose target is the reviewed product; a target already in its supplied history is excluded. The candidate set is the full set of items observed in the training file, minus products in that request’s prior history. Scoring that whole set rather than a sampled subset follows the guidance that testing should rank the candidates a system would actually serve [22]. A future target absent from the training catalog remains in the denominator and scores zero for every method.

For a ranked list R with one target y , NDCG@10 is $1/\log_2(r + 1)$ when y appears at one-based rank $r \leq 10$, and zero otherwise. We average this request-level value over the eligible cohort. This measures recovery of later positive reviews from the frozen training catalog. Reviews do not record which products a user saw, so the metric is not an exposure-adjusted engagement estimate.

3.2 Models and validation decision

The baseline scores each train-known item by either its lifetime count of positive training reviews or its count in the 365 days before the training cutoff. Counts are transformed by $\log(1 + x)$ and normalized by the largest count in that version of the model. Validation NDCG@10 chooses between these two controls. The co-review model takes each training user’s last 20 distinct positive items, counts shared users for item pairs, divides by the geometric mean of item frequencies, and keeps at most 50 neighbors per item. At a request, similarity contributions from the last 20 distinct history items are summed for eligible neighbors. The benchmark history includes prior reviews of any star rating, but this scorer ignores those ratings and therefore treats even a low-rated item as positive co-review evidence. The challenger uses $(1 - \alpha)$ times

the selected popularity score plus α times this co-review score, with $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$ selected on validation.

Before test rows are opened, a paired user-cluster bootstrap with 1,000 draws estimates the validation difference between the best challenger and its baseline. The route opens only if the lower endpoint of the percentile 95% interval is positive. Test reporting uses the saved selection and the same paired bootstrap procedure; the interval conditions on fitted models and one fixed test period. This gate is an operational rule, not a guarantee of future superiority. Sparse neighbor scoring produces the same top ten as scanning the train-known catalog because items without a neighbor adjustment retain their baseline order; exact-ranking tests check that property.

The method was developed on Video Games and fixed before the Musical Instruments test was opened. Video Games is therefore development evidence, while Musical Instruments is the fixed-method category check. Both categories come from one platform and one benchmark construction.

4 Results

Table 1: Full-catalog test quality. The paired interval is for hybrid minus the validation-selected popularity baseline. Video Games is exploratory; Musical Instruments is the fixed-method category check.

Category	Requests	Catalog	Popularity	Hybrid
Video Games	35,562	22,976	0.007151	0.010172
Musical Instruments	35,905	22,753	0.008007	0.008681

Both categories selected recent popularity and a hybrid weight of 0.75 on validation. In Musical Instruments, the 35,905 eligible test requests came from 11,648 users. The hybrid-minus-baseline NDCG@10 difference was 0.000674 [0.00040, 0.001338], and Recall@10 moved from 0.015903 to 0.017184. The interval excludes zero under the specified user resampling scheme, though its lower endpoint is close to zero. Video Games produced a larger exploratory difference of 0.003020 [0.002216, 0.003881] on 35,562 requests; that replay is not an independent confirmation. Figure 1 makes the difference in effect size and uncertainty visible without treating these comparisons as equivalent confirmation tests.

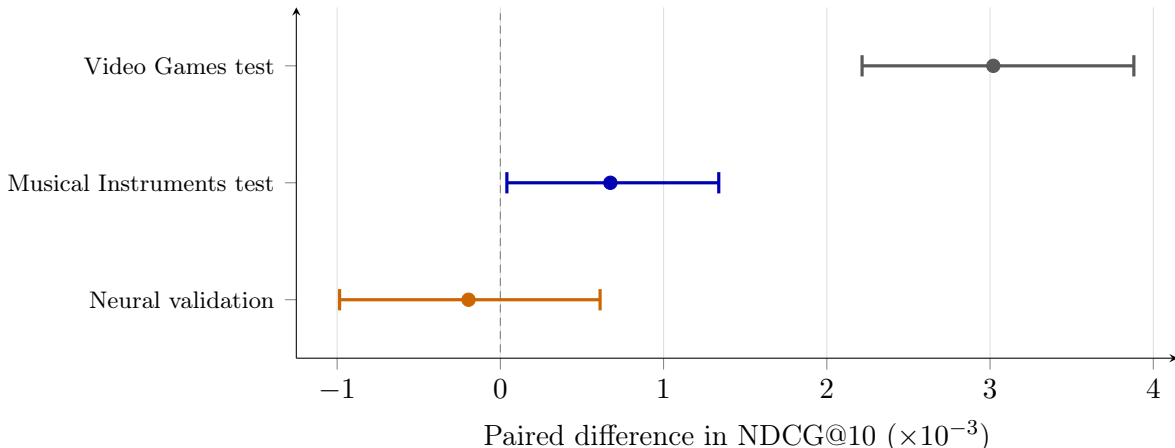


Figure 1: Paired NDCG@10 differences with user-cluster bootstrap 95% intervals. The two test rows compare hybrid with popularity; Musical Instruments is the fixed-method check and Video Games is exploratory. The later neural validation row compares the blended neural challenger with the active hybrid on Musical Instruments, not another test confirmation. Intervals condition on fitted models and one temporal period.

Catalog coverage helps explain the low absolute scores. In Musical Instruments, 12,305 eligible test targets were absent from the 22,753-item training catalog. Video Games had 22,051 such targets among 35,562 requests. Figure 2 shows how much of each test cohort is outside the frozen catalog. No reordering of train-known items can recover those requests. The measured hybrid gain thus sits inside a larger coverage limitation, and the test cannot establish that a user would engage with any recommended product.

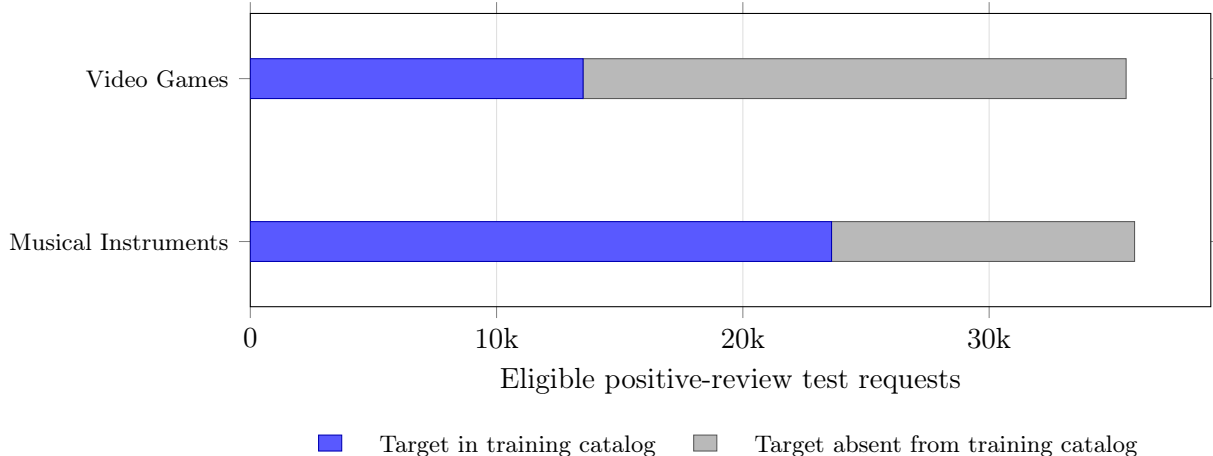


Figure 2: Frozen-catalog coverage of eligible test requests. The two segments sum to 35,905 Musical Instruments and 35,562 Video Games requests. Targets absent from the training catalog remain evaluation misses for every model; the chart does not imply those items were available to recommend at request time.

4.1 Neural challenger

The later experiment trains a two-tower retriever on 313,522 chronological pairs from positive Musical Instruments training reviews. A query tower pools embeddings of up to 20 prior distinct positive items and applies a small residual multilayer perceptron; an item tower embeds training products. Both towers produce normalized 64-dimensional vectors. Training uses three epochs of in-batch softmax at temperature 0.15, masking duplicate targets and products already known positive for a query so they are not treated as negatives. The learned model scores the train-known item array exactly, while items without positive training support are ineligible for its neural route. Requests without a supported history fall back to popularity.

On validation, the unblended neural model scored 0.001411 NDCG@10. Selecting a blend of 40% neural score and 60% recent popularity raised it to 0.011774, above recent popularity’s 0.010715 but below the active co-review hybrid’s 0.011969. The paired neural-minus-hybrid validation difference was -0.000195 $[-0.000985, 0.000611]$, so the stronger-challenger gate remained closed. On the already studied test period, the blend scored 0.008357 against 0.008681 for the hybrid. Since this model was designed after exposure to that period, the test comparison is exploratory and does not add a second fresh confirmation.

4.2 Evidence the frozen route discards

The route reads up to twenty distinct reviewed items at a request, regardless of their star ratings. It treats each as an attraction signal; the rating and review time are not used. An audit of the same records (`tools/signal_audit.py`) finds that 47,459 Musical Instruments validation history slots have a train-recorded rating below four; 35,444 of these fall inside the scorer’s twenty-item window. Another 96,984 slots lack a rating for that user–item pair in the training rows, but only 20,310 refer to products absent from the training catalog. Among distinct history items, 23% sit outside the window. For all 14,175

validation users who had a prior training record, the attached history reproduced the record exactly, so the window is a choice made by this scorer rather than a limit of the archive.

The shadow arms keep the trained co-review neighbors, candidate set, popularity control, and $\alpha = 0.75$ fixed. C1 subtracts γ times the neighbor score of history items whose train-recorded rating is at most three, with $\gamma \in \{0, 0.25, 0.5, 1\}$. C2 instead sets $\gamma = 0$ and weights train-recorded four-star and five-star items by 0.6 and 1.0, optionally multiplied by $2^{-d/\tau}$ for age d in days and half-life $\tau \in \{90, 365, 1095\}$; a missing train rating keeps unit attraction. C3 replaces the popularity term for requests with no usable co-review neighbors by the distribution of training users’ first positive reviews. C4 selects α separately for five history-length buckets. C5 asks about a fixed prefix of the 200 most often positively reviewed training products, excluding the current target and every item already in the supplied history; the observed archive rating is used as a hindsight answer. C6 projects training activity in Video Games through users shared with Musical Instruments, and fires only without usable local neighbors.

Table 2 scores these arms on 33,993 eligible Musical Instruments validation requests from 13,908 users against the active co-review hybrid at $\alpha = 0.75$. Configuration search and interval estimation use the same records. The intervals describe this cohort and do not provide an independent activation test.

Table 2: Shadow-profile arms on the Musical Instruments validation cohort, against the active hybrid at 0.011969 NDCG@10. These are exploratory validation results, not fresh confirmations. Intervals are paired user-cluster bootstrap estimates that do not account for selecting each configuration on the same cohort. C1 uses low ratings for repulsion; C2 instead grades and ages observed four- and five-star reviews. Their combination was not tested.

Arm	NDCG@10	Δ vs active	95% interval
C1 repulsion from sub-four ratings ($\gamma = 1$)	0.012418	+0.000449	[0.000266, 0.000626]
C2 star grading, 1095-day decay	0.012435	+0.000466	[0.000165, 0.000750]
C3 analogous first-positive-review prior	0.011824	−0.000145	[−0.000270, −0.000028]
C4 blend weight by history length	0.012523	+0.000554	[0.000062, 0.001057]
C5 oracle interview (100 prompts, $\gamma = 1$)	0.012840	+0.000871	[0.000537, 0.001172]
C6 adjacent surface, reachable probe	0.011960	−0.000009	[−0.000028, 0.000000]

C1 changes one term: a history item the platform observed below four now contributes repulsion as well as attraction. Setting $\gamma = 0$ reproduces the active route, and on real records that cell returns 0.011969, the active route’s own figure, so the parity check passes outside a unit test. The C2 ablation locates nearly all of that arm’s gain in the age of a record rather than in its star weight: a three-year half-life with flat rating weights moves NDCG@10 from 0.011969 to 0.012381, and grading on top of it adds 0.000054. C4’s value is the bucket table beneath it. Accuracy is not monotone in experience: it peaks at one or two recorded items (0.014825) and falls to 0.006784 for users with twenty or more, and where C4 differs from the frozen route it differs by *reducing* the personalised weight from 0.75 to 0.5. On this cohort, the frozen blend appears to put too much weight on long histories; the bucket weights require independent evaluation.

The questioning arm is a hindsight-oracle simulation: for each asked product, it reads that user’s recorded rating across train, validation, and test, while withholding the current target. A missing rating returns no answer. Thus 100 prompts per request yield about 0.227 recorded answers, or roughly one answer per 440 prompts; this is archive coverage under the simulation, not a real response rate. Its +0.000871 difference from the active route includes C1’s +0.000449 from already-held low ratings. The additional difference over C1 is +0.000422, and the current design cannot isolate the effect of newly elicited dislikes because the repulsion switch also activates the already-held ones. The likes-only 100-prompt cell gains +0.000461 over the active route, but that number also depends on this hindsight oracle.

C3 lost on the slice it touched rather than in aggregate. It fires on the 1,618 requests with no catalog-known history item, where its −0.000145 overall rescales to about −0.00305: substituting the aggregate first positive review of training users ranks the eventual target worse than the current crowd

does, a direction consistent with the MovieLens low-activity finding of Ji et al. [26]. Borrowing from an adjacent surface could not be measured as a comparison at all: the two category archives share 2,802 training users, roughly one in twenty, and the arm could reach a donor record for 47 of 33,993 requests (0.14%), below the 5% share the design set as the floor for reading it as a comparison.

4.3 Answering a sentence

The route reads one field of a request, so it cannot answer a request that states what the person wants. A final challenger supplies that capability: a typed sentence is matched against product text from the same catalogue, and its answer is reported beside the unchanged behavioural ranking rather than merged into one number. Product text is not part of the `timestamp_w_his` files, so it is fetched separately for items in the training catalog. For Musical Instruments, this yielded descriptions for all 22,753 products, titles for 22,751, and a recorded price for 14,927, together with earlier reviews from 55,981 people. Video Games has corresponding counts of 22,976 products, 22,975 titles, 14,919 prices, and 92,807 people with earlier reviews. An external query-to-product set contributes 157 in-catalogue Musical Instruments pairs and 271 Video Games pairs. Of these, 104 and 206 respectively come from observed ESCI shopping queries [16]; the remaining 53 and 65 come from Amazon-C4, whose queries were rewritten from reviews of the target product rather than observed in a search session [2].

Matching is BM25 written against the standard library (postings ordered by item ID, $k_1 = 1.2$, $b = 0.75$), with an optional dense route over the same documents from the all-MiniLM-L6-v2 sentence encoder (384 dimensions). The ranker uses whichever route is available, and the measured arm is scored on both: lexical in both categories and dense in Musical Instruments, where the embedding artifact exists. Negation is scoped to the words after a cue and ends at the next cue, so a phrase can rule out a product class. A stated figure is treated as a limit on the served route: products recorded above it are not offered, and products with no recorded price are kept.

Six wording conditions are scored against the route validation already chose, under the same paired gate. Two are read back from the person’s own earlier reviews, one raw and one scrubbed to product words appearing in under 1% of listings, so that shared product vocabulary cannot carry the answer by itself. One takes wording from a different reviewer of the same product, which is the generalisation control. The remaining conditions attach text using the eventual target: its exact title, a randomly drawn title, or one of the external query pairs. ESCI queries were observed in shopping search, but were not observed from these review users at these request times; C4 queries are review-derived rewrites. All three conditions are therefore target-conditioned diagnostics, not evaluations of a live request’s phrasing. Table 3 reports how often each condition’s answer list contained the eventual target, beside the ceiling a confined phrase works against.

The arm does not clear the gate. On the full Musical Instruments test split the confined configuration (weight 0.25, repulsion 0, price penalty 0.5) loses 0.001678 NDCG@10 against the same frozen base (95% interval $[-0.002318, -0.000980]$), and on the exploratory Video Games split it loses 0.002858 (95% interval $[-0.003598, -0.002195]$), so the confined arm activates in neither category. On the validation cohort from which the configuration was selected the loss is smaller (-0.001057 , interval $[-0.002147, -0.000042]$). The dense route that re-scores the same shortlist with all-MiniLM-L6-v2 embeddings is directionally positive there ($+0.001048$, interval $[-0.000062, +0.002109]$) but does not open the gate. That direction does not hold on the full Musical Instruments test split, where the same confined configuration loses 0.001282 (95% interval $[-0.001944, -0.000577]$).

Among the phrase-like conditions, only the mixed external-pair condition has a positive full-split paired difference: $+0.002394$ for lexical and $+0.002132$ for encoded Musical Instruments (intervals $[0.001892, 0.002875]$ and $[0.001669, 0.002622]$). These averages include all 35,905 test requests, although an external pair attaches to only 681; the pairs are selected by known target, not observed from those users at request time. On Video Games, the corresponding lexical difference is $+0.003115$ across all 35,562 requests, with a pair attached to 824. A source audit of the fixed lexical run attributes the Musical

Table 3: Wording reach on the full test splits (35,905 Musical Instruments requests and 35,562 Video Games requests). Each condition cell gives the number of requests whose answer list contained the eventual target out of the requests where the condition had wording to read, with the fraction in parentheses; the answer list is read without the confinement that produces the measured number, so reach is a diagnostic, not a score. The last row is not a wording condition: it is the share of requests whose target sat inside the top-200 shortlist a confined phrase was allowed to promote at all.

Wording condition	Musical Instruments	Video Games
Own earlier reviews	627 / 25,926 (2.42%)	390 / 22,320 (1.75%)
Own reviews, scrubbed to rare product words	584 / 25,926 (2.25%)	366 / 22,320 (1.64%)
Another reviewer of the same product	193 / 25,926 (0.74%)	124 / 22,320 (0.56%)
Target product title	23,596 / 23,600 (99.98%)	13,507 / 13,511 (99.97%)
Randomly drawn product title	263 / 35,905 (0.73%)	150 / 35,560 (0.42%)
External pairs (ESCI + C4)	325 / 681 (47.72%)	276 / 824 (33.50%)
Target inside the promotable shortlist	11,925 / 35,905 (33.21%)	8,267 / 35,562 (23.25%)

Instruments difference to +0.002538 from C4 pairs on 499 requests and -0.000144 from ESCI pairs on 182 requests. In Video Games the corresponding full-split contributions are +0.002690 on 616 C4-linked requests and +0.000425 on 208 ESCI-linked requests. These contributions are descriptive, and repeated requests share target-linked pairs; the positive mixed interval cannot establish a benefit from observed user queries. Figure 3 shows the three own-words regressions and their paired intervals.

Table 3 separates two constraints. The target lay inside the promotable shortlist for 33.21% of Musical Instruments requests, while unconstrained lexical search using earlier review prose retrieved it for 2.42% of requests with usable wording. These are marginal rates over different denominators; their product is not an observed joint ceiling. An exact target title retrieves its own product 99.98% of the time, confirming that the index handles target-informed wording, while the mixed external pairs retrieve the target in 47.72% of their matched sample. Thus both the prior-review proxy and shortlist coverage constrain this arm; the title probe does not rule out retrieval failures for natural queries. The serving configuration therefore differs from the measured one on purpose: the served route lets a phrase reach the whole catalogue and keeps the behavioural ranking answering whenever the sentence is empty, while the arm that produces a number keeps the phrase confined and reports the loss.

The capability is cheap to run. Loading 22,753 product descriptions and building the index takes 3.1 to 3.2 s at startup, a phrased request is answered in 38.3 ms at the median and 48.1 ms at the 95th percentile over 200 scored Musical Instruments requests (21.2 ms and 29.3 ms for Video Games) in process, the dense route answers the same 200 Musical Instruments requests in 26.9 ms at the median and 62.2 ms at the 95th percentile, and the service holds up to eight earlier sentences in memory so it can re-answer from something already said without being asked again. The dense route, when the optional dependency is installed, encodes the same catalogue in 37.4 s on one consumer GPU.

5 Guarded service and reproducibility

The public implementation serves baseline, co-review shadow, optional neural shadow, active ranking, and an explicit fallback reason through a loopback HTTP API. A saved validation decision determines the active route. Empty or unsupported histories and challenger failures return the popularity baseline. A local model bundle checks its source hash before serving; optional neural weights and their training source are checked as well. A phrased request is served by a separate route whose answer is returned beside the behavioural one, with the terms it looked for, the terms it ruled out, and any stated figure reported back, so a reader can see which part of an answer came from words. Up to eight sentences are held in process memory and cleared when the process stops; submitted histories are not written to disk. A synthetic

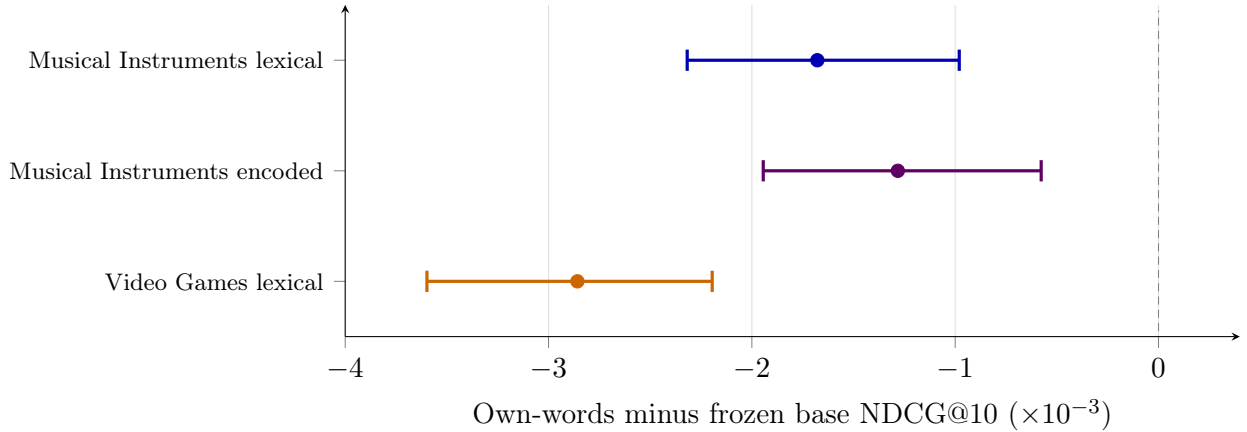


Figure 3: Exploratory own-words wording arm against the frozen behavioural base on the full test splits. Points are paired NDCG@10 differences; lines are user-cluster bootstrap 95% intervals. Lexical retrieval was measured in both categories; encoded retrieval was measured only in Musical Instruments. Both test periods had already been opened by earlier work, so these intervals describe regressions on those cohorts rather than independent confirmation.

browser walkthrough illustrates these decisions with invented users and product names. Its controls do not alter the measured results.

The measured in-process Python ranking p95 was 0.611 ms for the Video Games replay and 0.760 ms for Musical Instruments, including baseline and co-review challenger scoring but excluding HTTP, loading, concurrency, and network time. The neural batch-ranking timing has a different scope and is not a like-for-like service latency comparison. The repository contains source, evaluation methods, aggregate JSON results, and tests at <https://github.com/ahnafnafee/recommendation-decision-lab>. Downloaded review files, row-level predictions, and fitted weights are not redistributed. The dataset remains available from its owner subject to the owner’s terms [1].

6 Limitations and conclusion

The 5-core subset was formed using support across the full corpus, which retrospectively selects users and products. Review activity is not an impression, purchase, or unbiased preference label. Our user-cluster intervals hold the fitted models and temporal period fixed and do not account for shared-item dependence, model refitting, selection across methods, or future regimes. The second category tests a different product domain within the same platform, not transfer to a new service. The neural route was assessed after the category test was known, and one compact architecture does not exhaust learned retrieval methods.

The profile arms were scored on the validation cohort alone, because both test periods in this study had already been opened by earlier work. Their positive intervals were computed after selecting configurations on the same validation cohort; they motivate a fresh evaluation but do not justify activating a new route. The arms were not fitted together, and their interaction is unknown. The wording arm carries further limits. Its primary condition reads earlier reviews about other products, not a statement of intent for the current request. The scrubbed variant removes rare words from the known target’s title, but its small reach change does not identify the cause of the proxy’s failure. External pairs are attached through the known answer and mix observed ESCI queries with C4 rewrites, so the mixed interval does not validate a live-query intervention. The title conditions are target-informed probes, not phrases observed at the review request. Product text is fetched by whoever runs the code and is not redistributed here, and the encoded route was measured only on Musical Instruments, where the embedding artifact exists.

Within those limits, the fixed co-review hybrid beat a validation-selected popularity baseline by a

small margin in Musical Instruments. The later neural blend improved on popularity but did not earn the active route. Applying the same rule to the evidence the route leaves unread is more informative than either result on its own: two arms using already-held information had positive descriptive validation differences, while the simulated questioning arm needed about 440 prompts per recorded answer under its hindsight oracle, a prior derived from comparable users was declined, and the arm that answers a sentence was declined on the shared metric while remaining the only one that changes what the system can respond to. Together, the experiments show why a recommendation system should report the baseline it would actually use, evaluate the catalog it can actually retrieve, look twice at the records it already received, and retain a clear fallback when additional capacity or additional evidence lacks sufficient proof.

References

- [1] McAuley Lab. *Amazon Reviews’23: Pure IDs (5-Core)*. https://amazon-reviews-2023.github.io/data_processing/5core.html.
- [2] Y. Hou, J. Li, X. Fu, Z. He, A. Yan, X. Chen, and J. McAuley. Bridging Language and Items for Retrieval and Recommendation: Benchmarking LLMs as Semantic Encoders. *arXiv:2403.03952*, version 2, 2026. <https://arxiv.org/abs/2403.03952>.
- [3] Y. Ji, A. Sun, J. Zhang, and C. Li. A Critical Study on Data Leakage in Recommender System Offline Evaluation. *ACM Transactions on Information Systems*, 41(3), Article 75, 2023. <https://doi.org/10.1145/3569930>.
- [4] D. Gusak, A. Volodkevich, A. Klenitskiy, A. Vasilev, and E. Frolov. Time to Split: Exploring Data Splitting Strategies for Offline Evaluation of Sequential Recommenders. In *Proceedings of RecSys ’25*, pp. 874–883, 2025. <https://doi.org/10.1145/3705328.3748164>.
- [5] P. Covington, J. Adams, and E. Sargin. Deep Neural Networks for YouTube Recommendations. In *Proceedings of RecSys ’16*, pp. 191–198, 2016. <https://doi.org/10.1145/2959100.2959190>.
- [6] J. Wang, J. Zhu, and X. He. Cross-Batch Negative Sampling for Training Two-Tower Recommenders. In *Proceedings of SIGIR ’21*, pp. 1632–1636, 2021. <https://doi.org/10.1145/3404835.3463032>.
- [7] A. M. Rashid, G. Karypis, and J. Riedl. Learning preferences of new users in recommender systems: an information theoretic approach. *SIGKDD Explorations*, 10(2), pp. 90–100, 2008. <https://doi.org/10.1145/1540276.1540302>.
- [8] D. Crankshaw, X. Wang, G. Zhou, M. J. Franklin, J. E. Gonzalez, and I. Stoica. Clipper: A Low-Latency Online Prediction Serving System. *arXiv:1612.03079*, 2016 (later *MLSys ’18*). <https://arxiv.org/abs/1612.03079>.
- [9] K. Hendrickx, L. Perini, D. Van der Plas, W. Meert, and J. Davis. Machine Learning with a Reject Option: A survey. *Machine Learning*, 113(5), pp. 3073–3110, 2024. <https://doi.org/10.1007/s10994-024-06534-x>.
- [10] P. Hemmer, L. Thede, M. Vössing, J. Jakubik, and N. Kühl. Learning to Defer with Limited Expert Predictions. In *Proceedings of AAAI-23*, 37(5), pp. 6002–6011, 2023. <https://doi.org/10.1609/aaai.v37i5.25742>.
- [11] K. Krauth, S. Dean, A. Zhao, W. Guo, M. Curmei, B. Recht, and M. I. Jordan. Do Offline Metrics Predict Online Performance in Recommender Systems? *arXiv:2011.07931*, 2020. <https://arxiv.org/abs/2011.07931>.

- [12] A. Gilotte, C. Calauzènes, T. Nedelec, A. Abraham, and S. Dollé. Offline A/B testing for Recommender Systems. In *Proceedings of KDD '18*, 2018. <https://arxiv.org/abs/1801.07030>.
- [13] C.-W. Hsu, M. Mladenov, O. Meshi, J. Pine, H. Pham, S. Li, X. Liang, A. Polishko, L. Yang, B. Scheetz, and C. Boutilier. Minimizing Live Experiments in Recommender Systems: User Simulation to Evaluate Preference Elicitation Policies. *arXiv:2409.17436*, 2024. <https://arxiv.org/abs/2409.17436>.
- [14] S. Gupta, H. Oosterhuis, and M. de Rijke. A First Look at Selection Bias in Preference Elicitation for Recommendation. In *CONSEQUENCES'23 workshop at RecSys '23*, 2023. <https://arxiv.org/abs/2405.00554>.
- [15] J. Huang, A. Sharma, S. Sun, L. Xia, D. Zhang, P. Pronin, J. Padmanabhan, G. Ottaviano, and L. Yang. Embedding-based Retrieval in Facebook Search. In *Proceedings of KDD '20*, 2020. <https://arxiv.org/abs/2006.11632>.
- [16] C. K. Reddy, L. Márquez, F. Valero, N. Rao, H. Zaragoza, S. Bandyopadhyay, A. Biswas, A. Xing, and K. Subbian. Shopping Queries Dataset: A Large-Scale ESCI Benchmark for Improving Product Search. *arXiv:2206.06588*, 2022. <https://arxiv.org/abs/2206.06588>.
- [17] N. Reimers and I. Gurevych. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of EMNLP 2019*, 2019. <https://arxiv.org/abs/1908.10084>.
- [18] H. A. Rahmani, N. Craswell, E. Yilmaz, B. Mitra, and D. Campos. Synthetic Test Collections for Retrieval Evaluation. In *Proceedings of SIGIR 2024*, 2024. <https://arxiv.org/abs/2405.07767>.
- [19] H. A. Rahmani, V. Ramineni, E. Yilmaz, N. Craswell, and B. Mitra. Towards Understanding Bias in Synthetic Data for Evaluation. In *Proceedings of CIKM 2025*, 2025. <https://arxiv.org/abs/2506.10301>.
- [20] Z. Hussain and K. Nielbo. The Coverage Illusion: From Pre-retrieval Routing Failure to Post-retrieval Cascades in a Production RAG System. *arXiv:2605.27220*, 2026. <https://arxiv.org/abs/2605.27220>.
- [21] Z. Meng, R. McCreadie, C. Macdonald, and I. Ounis. Exploring Data Splitting Strategies for the Evaluation of Recommendation Models. *arXiv:2007.13237*, 2020. <https://arxiv.org/abs/2007.13237>.
- [22] B. Hidasi and Á. T. Czapp. Widespread Flaws in Offline Evaluation of Recommender Systems. In *Proceedings of the 17th ACM Conference on Recommender Systems*, 2023. <https://arxiv.org/abs/2307.14951>.
- [23] T. Schnabel. Where Do We Go From Here? Guidelines For Offline Recommender Evaluation. *arXiv:2211.01261*, 2022. <https://arxiv.org/abs/2211.01261>.
- [24] Yunzhu Pan, Chen Gao, Jianxin Chang, Yanan Niu, Yang Song, Kun Gai, Depeng Jin, and Yong Li. Understanding and Modeling Passive-Negative Feedback for Short-video Sequential Recommendation. In *Proceedings of the 17th ACM Conference on Recommender Systems*, 2023. *arXiv:2308.04086*. <https://arxiv.org/abs/2308.04086>.
- [25] Yueqi Wang, Yoni Halpern, Shuo Chang, Jingchen Feng, Elaine Ya Le, Longfei Li, Xujian Liang, Min-Cheng Huang, Shane Li, Alex Beutel, Yaping Zhang, and Shuchao Bi. Learning from Negative User Feedback and Measuring Responsiveness for Sequential Recommenders. In *Proceedings of the 17th ACM Conference on Recommender Systems, Industry Track*, 2023. *arXiv:2308.12256*. <https://arxiv.org/abs/2308.12256>.

- [26] Yitong Ji, Aixin Sun, Jie Zhang, and Chenliang Li. A Re-visit of the Popularity Baseline in Recommender Systems. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020. arXiv:2005.13829. <https://arxiv.org/abs/2005.13829>.